

摘要：

瑞穗最新调研强烈看多博通，预测2028年谷歌TPU出货将达炸裂的3500万颗。由于博通是谷歌TPU的ASIC设计合作方，TPU出货量直接对应其业务收入。但该数据高达全行业总测量的2.3倍，因严重脱离供应链现实遭业内质疑：莫不是实习生用AI跑出的假结果？

博通股价承压之际，近日一份流传出的瑞穗（Mizuho）渠道调研显示，该行对博通发出强烈看多信号。核心数据只有一个，但足够炸裂：**预测2028年谷歌TPU出货量将超过3500万颗**。

这个数字意味着什么？2025年谷歌TPU出货量约240万颗，2026年预估约430万颗，到2028年跳到3500万颗——**两年增长约8倍，隐含复合年增速约185%**。

谷歌的TPU是定制AI芯片（ASIC），博通是其核心设计合作方。TPU每多出货一颗，博通的ASIC业务就多一份收入。简单说，谷歌出需求和架构，博通负责芯片设计服务和封装，再交给台积电代工。每一颗TPU出货，博通都有对应的设计服务收入。

不过，该数字的数字可信度遭到市场观察员直接质疑：**全行业2028年ASIC总量预测才1500万，谷歌一家就要出2.3倍？“我真的很好奇，是不是瑞穗某个实习生让Claude外推了一条趋势线，然后没人去核对输出结果。”**



显示翻译

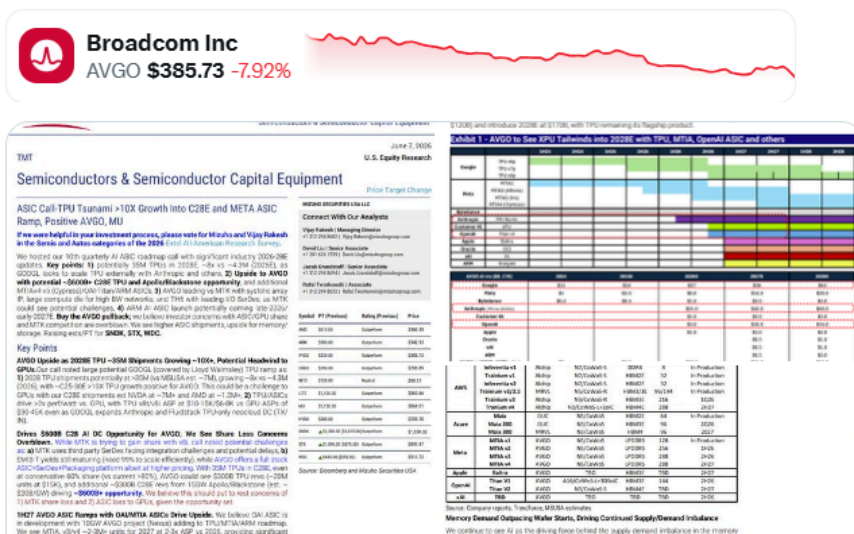
Mizuho ASIC channel checks: pounding table on \$AVGO

"potential >35M TPU shipments in 2028E (up 8x from 2026E ~4.3M and 2025 ~2.4M), which is significantly above our estimates of ~7M ASIC shipments for AVGO"

"We believe OAI ASIC is in development with 10GW AVGO project (Nexus) adding to TPU/MTIA/ARM roadmap."

"Our call noted \$ARM AI ASIC chip development ongoing as we continue to expect a launch in late-2026E/early-2027E"

"\$MU : With potential for TPU shipments to reach a cumulative 50M units in 2026-28E, we think ASICs could command an increasing share of the HBM market, especially as some of the next-gen ASICs move to HBM4e"



3500万颗，这个数字有多离谱

对比一下几个参照系：

- **Counterpoint Research**预测，到2028年，谷歌、亚马逊、微软、Meta等所有超大规模云厂商的ASIC出货量加在一起，总计约1500万颗。
- **Wolfe Research**的渠道调研显示，谷歌TPU出货量约700万颗。
- **瑞穗自己的模型**此前预测ASIC出货量也是约700万颗。

换句话说，按瑞穗渠道调研的数字，谷歌一家公司2028年的TPU出货量，将超过地球上所有ASIC项目加起来的两倍。

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <div> <div>Industry Report</div> <div>Semiconductors &amp; Semiconductor Capital Equipment</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |            |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <div> <div>June 7, 2026</div> <div>U.S. Equity Research</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |            |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| <div> <div>TMT</div> <div>Semiconductors &amp; Semiconductor Capital Equipment</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div> <div>Price Target Change</div> <div> <div> <div>MIZUHO SECURITIES USA LLC</div> <div>Connect With Our Analysts</div> <div> <div>Vijay Rakesh   Managing Director</div> <div>+1 312 294 8682   Vijay.Rakesh@mizuhogroup.com</div> </div> <div> <div>David Liu   Senior Associate</div> <div>+1 201 626 1539   David.Liu@mizuhogroup.com</div> </div> <div> <div>Jacob Grandstaff   Senior Associate</div> <div>+1 312 294 8694   Jacob.Grandstaff@mizuhogroup.com</div> </div> <div> <div>Rafal Tworkowski   Associate</div> <div>+1 312 294 8632   Rafal.Tworkowski@mizuhogroup.com</div> </div> </div> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                      |                   |            |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| <div> <div>ASIC Call-TPU Tsunami &gt;10X Growth Into C28E and META ASIC Ramp, Positive AVGO, MU</div> <div> <div>If we were helpful in your investment process, please vote for Mizuho and Vijay Rakesh in the Semis and Autos categories of the 2026 Extel All-American Research Survey.</div> <div> <div>We hosted our 10th quarterly AI ASIC roadmap call with significant industry 2026-28E updates. <b>Key points:</b> <b>1)</b> potentially 35M TPUs in 2028E, ~8x vs ~4.3M (2026E), as GOOGL looks to scale TPU externally with Anthropic and others, <b>2) Upside to AVGO with potential ~\$600B+ C28E TPU and Apollo/Blackstone opportunity</b>, and additional MTIAv4-v5 (Cypress)/OAI-Titan/ARM ASICs, <b>3)</b> AVGO leading vs MTK with systolic array IP, large compute die for high BW networks, and TH6 with leading I/O SerDes, as MTK could see potential challenges, <b>4)</b> ARM AI ASIC launch potentially coming late-2026/early-2027E. <b>Buy the AVGO pullback;</b> we believe investor concerns with ASIC/GPU share and MTK competition are overblown. We see higher ASIC shipments, upside for memory/storage. Raising ests/PT for <b>SNDK, STX, WDC</b>.</div> </div> </div> </div> | <table> <tr> <th>Symbol</th><th>PT (Previous)</th><th>Rating (Previous)</th><th>Price</th></tr> <tr> <td>AMD</td><td>\$615.00</td><td>Outperform</td><td>\$466.38</td></tr> <tr> <td>ARM</td><td>\$500.00</td><td>Outperform</td><td>\$342.93</td></tr> <tr> <td>AVGO</td><td>\$530.00</td><td>Outperform</td><td>\$385.73</td></tr> <tr> <td>ORDO</td><td>\$290.00</td><td>Outperform</td><td>\$206.89</td></tr> <tr> <td>INTC</td><td>\$128.00</td><td>Neutral</td><td>\$99.17</td></tr> <tr> <td>LITE</td><td>\$1,100.00</td><td>Outperform</td><td>\$863.66</td></tr> <tr> <td>MU</td><td>\$1,150.00</td><td>Outperform</td><td>\$864.01</td></tr> <tr> <td>NVDA</td><td>\$300.00</td><td>Outperform</td><td>\$205.10</td></tr> <tr> <td>SNDK</td><td>▲ \$2,200.00 (\$1,825.00)</td><td>Outperform</td><td>\$1,559.32</td></tr> <tr> <td>STX</td><td>▲ \$1,090.00 (\$875.00)</td><td>Outperform</td><td>\$847.47</td></tr> <tr> <td>WDC</td><td>▲ \$685.00 (\$550.00)</td><td>Outperform</td><td>\$511.72</td></tr> </table> |                   | Symbol     | PT (Previous) | Rating (Previous) | Price | AMD | \$615.00 | Outperform | \$466.38 | ARM | \$500.00 | Outperform | \$342.93 | AVGO | \$530.00 | Outperform | \$385.73 | ORDO | \$290.00 | Outperform | \$206.89 | INTC | \$128.00 | Neutral | \$99.17 | LITE | \$1,100.00 | Outperform | \$863.66 | MU | \$1,150.00 | Outperform | \$864.01 | NVDA | \$300.00 | Outperform | \$205.10 | SNDK | ▲ \$2,200.00 (\$1,825.00) | Outperform | \$1,559.32 | STX | ▲ \$1,090.00 (\$875.00) | Outperform | \$847.47 | WDC | ▲ \$685.00 (\$550.00) | Outperform | \$511.72 |
| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PT (Previous)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rating (Previous) | Price      |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| AMD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | \$615.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Outperform        | \$466.38   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| ARM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | \$500.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Outperform        | \$342.93   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| AVGO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | \$530.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Outperform        | \$385.73   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| ORDO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | \$290.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Outperform        | \$206.89   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| INTC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | \$128.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Neutral           | \$99.17    |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| LITE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | \$1,100.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outperform        | \$863.66   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| MU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | \$1,150.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Outperform        | \$864.01   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| NVDA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | \$300.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Outperform        | \$205.10   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| SNDK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ▲ \$2,200.00 (\$1,825.00)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Outperform        | \$1,559.32 |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| STX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ▲ \$1,090.00 (\$875.00)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Outperform        | \$847.47   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| WDC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ▲ \$685.00 (\$550.00)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Outperform        | \$511.72   |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| <div> <div>Key Points</div> <div> <div>AVGO Upside as 2028E TPU ~35M Shipments Growing ~10X+, Potential Headwind to GPUs.</div> <div>Our call noted large potential GOOGL (covered by Lloyd Walmsley) TPU ramp as: <b>1)</b> 2028 TPU shipments potentially at &gt;35M (vs MSUSA est. ~7M), growing ~8x vs ~4.3M (2026), with ~C25-30E &gt;10X TPU growth positive for AVGO. This could be a challenge to GPUs with our C28E shipments est NVDA at ~7M+ and AMD at ~1.2M+, <b>2)</b> TPU/ASICs drive &gt;2x perf/watt vs. GPU, with TPU v8t/v8i ASP at \$10-15K/\$6-8K vs GPU ASPs of \$30-45K even as GOOGL expands Anthropic and Fluidstack TPU-only neocloud DC (TX/IN).</div> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <div> <div>Source: Bloomberg and Mizuho Securities USA</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |            |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| <div> <div>Drives \$600B C28 AI DC Opportunity for AVGO; We See Share Loss Concerns Overblown.</div> <div>While MTK is trying to gain share with v8i, call noted potential challenges as: <b>a)</b> MTK uses third party SerDes facing integration challenges and potential delays, <b>b)</b> EMIB-T yields still maturing (need 99% to scale efficiently), while AVGO offers a full stack ASIC+SerDes+Packaging platform albeit at higher pricing. With 35M TPUs in C28E, even at conservative 60% share (vs current &gt;80%), AVGO could see \$300B TPU revs (~20M units at \$15K), and additional ~\$300B C28E revs from 15GW Apollo/Blackstone (est. ~\$20B/GW) driving ~<b>\$600B+ opportunity</b>. We believe this should put to rest concerns of <b>1)</b> MTK share loss and <b>2)</b> ASIC loss to GPUs, given the opportunity set.</div> </div>                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |            |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| <div> <div>1H27 AVGO ASIC Ramps with OAI/MTIA ASICs Drive Upside.</div> <div>We believe OAI ASIC is in development with 10GW AVGO project (Nexus) adding to TPU/MTIA/ARM roadmap. We see MTIA, v3/v4 ~2-3M+ units for 2027 at 2-3x ASP vs 2026, providing significant tailwind. We see strong AVGO competitive moats with: <b>1)</b> critical logic die systolic array IP with large compute die, <b>2)</b> leadership in high BW scaled training/networking integration with TH6 and industry-leading SerDes I/O, posing challenges to competitors such as MTK (2454TW, covered by Kevin Wang-HK).</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |            |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |
| <div> <div>ARM AI ASIC Offering Potentially Late-C26E/Early-C27E.</div> <div>Our call noted ARM AI ASIC chip development ongoing as we continue to expect a launch in late-2026E/early-2027E. We see it significantly expanding ARM's AI DC opportunity (with our upside <b>ARM-SOTP</b>), adding to its IP/AGI CPU, with standalone AI ASIC TAM expected at &gt;\$1T by ~2030E. We believe the upside scenario for TPU ASICs for 2028 and upcoming OAI/MTIA/ARM ASICs are all positive for memory (MU/SNDK) and storage (WDC/STX) attach.</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |            |               |                   |       |     |          |            |          |     |          |            |          |      |          |            |          |      |          |            |          |      |          |         |         |      |            |            |          |    |            |            |          |      |          |            |          |      |                           |            |            |     |                         |            |          |     |                       |            |          |

市场观察员@LinQingV在X平台直接点名质疑这份调研。



“你们有没有认真想过，2028年出货3500万颗TPU意味着什么？”他写道。

他的质疑逻辑很清晰：从430万到3500万，两年复合增速约185%，“就算加上联发科双供应商和Anthropic 3.5GW的协议，全球没有任何一条供应链能以这种速度扩张。”

他也注意到，瑞穗给出的美光HBM需求测算在数字上是自治的——2026年至2028年累计TPU出货量：430万+1070万+3500万=5000万颗，数学没错。但他的结论是：“内部自治”和“物理上可行”是两回事。

他最后写道：“我真的很好奇，是不是瑞穗某个实习生让Claude外推了一条趋势线，然后没人去核对输出结果。”



Macro Lin | 市场观察员  
@LinQingV

订阅



显示翻译

Have you actually thought about what 35M TPU shipments in 2028 means?

Counterpoint estimates total ASIC shipments across ALL hyperscalers at 15M by 2028. Wolfe's channel checks say ~7M TPU. Mizuho's own model says ~7M.

So their channel check number is 5x their own estimate and 2.3x the entire industry total. That's Google alone outshipping every ASIC program on earth combined, twice over.

Going from 4.3M to 35M in two years implies ~185% CAGR. Even with MediaTek dual-sourcing and the Anthropic 3.5GW deal, no supply chain on the planet scales like that.

The 50M cumulative figure for MU/HBM is internally consistent (4.3+10.7+35=50), so someone did the math. But “internally consistent” and “physically possible” are two different things.

Genuinely curious if a Mizuho intern just asked Claude to extrapolate a trendline and nobody checked the output.



Sean @sean\_\_\_\_\_ · 12小时

Mizuho ASIC channel checks: pounding table on \$AVGO  
"potential >35M TPU shipments in 2028E (up 8x from 2026E ~4.3M and 2025 ~2.4M), which is significantly above our estimates of ~7M ASIC shipments for AVGO"

## 除谷歌TPU之外，瑞穗还提到了什么

这份渠道调研并不只涉及谷歌。

瑞穗同时提到，OpenAI的ASIC芯片正在开发中，与博通的"Nexus"项目相关，该项目规模达10GW，将补充谷歌TPU、Meta MTIA以及ARM芯片的路线图。

关于ARM，测算显示其AI ASIC芯片开发正在推进，预计将于2026年底至2027年初发布。

对美光的影响也被单独提及：如果2026年至2028年TPU累计出货量达到5000万颗，ASIC将在HBM市场中占据越来越大的份额，尤其是随着下一代ASIC转向HBM4e规格。



Potential Significant TPU Acceleration with 2028E Growing 8-10X vs 2026E, Driving UPSIDE to AVGO

Our call noted a potential >35M TPU shipments in 2028E (up 8x from 2026E ~4.3M and 2025 ~2.4M), which is significantly above our estimates of ~7M ASIC shipments for AVGO, implying significant upside. Additionally, would note that AVGO could see 50M total TPU shipments from 2026-28E, which provide some upside alongside accelerating demand for MTIA v3/4/5 and OpenAI's upcoming ASIC, which AVGO will be a key partner. Given the strength around upcoming ASIC ramps, we are raising our AVGO AI revenue by customer estimates to 2027E at \$122B (prior \$120B) and introduce 2028E at \$170B, with TPU remaining its flagship product.

Exhibit 1 - AVGO to See XPU Tailwinds into 2028E with TPU, MTIA, OpenAI ASIC and others

|             |                 | 1H24 | 2H24 | 1H25 | 2H25 | 1H26 | 2H26 | 1H27 | 2H27 | 1H28 | 2H28 |
|-------------|-----------------|------|------|------|------|------|------|------|------|------|------|
| Google      | TPU v5p         |      |      |      |      |      |      |      |      |      |      |
|             | TPU v7p         |      |      |      |      |      |      |      |      |      |      |
|             | TPU v8p         |      |      |      |      |      |      |      |      |      |      |
| Meta        | MTIA1           |      |      |      |      |      |      |      |      |      |      |
|             | MTIA2 (Athena)  |      |      |      |      |      |      |      |      |      |      |
|             | MTIA3 (Iris)    |      |      |      |      |      |      |      |      |      |      |
|             | MTIA4 (Olympus) |      |      |      |      |      |      |      |      |      |      |
| Bytedance   |                 |      |      |      |      |      |      |      |      |      |      |
| Anthropic   | TPU Racks       |      |      |      |      |      |      |      |      |      |      |
| Customer #5 | XPU             |      |      |      |      |      |      |      |      |      |      |
| OpenAI      | Titan v1        |      |      |      |      |      |      |      |      |      |      |
| Apple       | Baltira         |      |      |      |      |      |      |      |      |      |      |
| Oracle      | OCI             |      |      |      |      |      |      |      |      |      |      |
| xAI         | X1              |      |      |      |      |      |      |      |      |      |      |
| ARM         | Stargate        |      |      |      |      |      |      |      |      |      |      |

| AVGO AI rev (\$B, C/E)       | 2024  | 2025E  | 2026E  | 2027E   | 2028E   |
|------------------------------|-------|--------|--------|---------|---------|
| Google                       | \$11  | \$14   | \$27   | \$35    | \$54    |
| Meta                         | \$1   | \$3.0  | \$6.0  | \$14.0  | \$20.0  |
| Bytedance                    | \$0.2 | \$0.5  | \$1.0  | \$2.0   | \$3.0   |
| Anthropic (TPU ex-GOOL)      |       |        | \$21.0 | \$42.0  | \$50.0  |
| Customer #5                  |       |        | \$1.0  | \$2.0   | \$3.0   |
| OpenAI                       |       |        | \$3.0  | \$10.0  | \$15.0  |
| Apple                        |       |        | \$1.0  | \$1.0   | \$2.0   |
| Oracle                       |       |        |        | \$0.5   | \$1.0   |
| xAI                          |       |        |        | \$0.5   | \$1.0   |
| ARM                          |       |        |        | \$0.5   | \$3.0   |
| All Other (MSFT, AMZN, etc.) | \$2   | \$7.0  | \$10.0 | \$15.0  | \$18.0  |
| Total (est range +/-)        | \$14  | \$24.5 | \$70.0 | \$122.0 | \$170.0 |

Source: Company reports, MSUSA estimates

ASICs to Drive HBM Demand Alongside GPU Ramps

With potential for TPU shipments to reach a cumulative 50M units in 2026-28E, we think ASICs could command an increasing share of the HBM market, especially as some of the next-gen ASICs move to HBM4e. We continue to see the HBM market reaching \$246B by 2028E, driven by increasing capacity per GPU/ASIC, as well as pricing tailwinds with HBM4/4e potentially carrying strong premiums over their predecessors. We note that many ASIC suppliers are likely to skip the HBM4 generation and go straight to HBM4e, allowing them to take advantage of some the more easily available HBM3e capacity in 2H26/2027E, while preparing for increased customization on HBM4e to drive better performance.

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